Counting Fruit a Camera Cannot See: Estimating Mango Load from Few Viewpoints

CREST Gold Award: project report.

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Abstract

My project asks how many mangoes a tree carries when a camera can only see some of them. Fruit detectors already find visible mangoes almost perfectly, so the remaining error is the fruit hidden behind leaves, limbs and other fruit, which published work corrects with a multiplier fitted per orchard. I built a canopy simulator whose fruit count is known by construction, and an estimator that treats each camera position as a survey occasion and uses the reconstructed canopy to measure the volume no camera could see into. On sixty simulated trees the estimator reached a mean absolute error of 3.5% in total count from three viewpoints, against 7.9% for the fitted multiplier and 24.5% for counting what is found. Three viewpoints with it matched twelve without. It fails in one specific place, on two viewpoints with a poor detector, where a blunt constant beats it. The measurement that would confirm any of this on real fruit is a harvest away.

Contents

1. Introduction

My family buys Alphonso every summer from a grower in Ratnagiri. Two summers ago he told my father he had hired eleven people for a picking he thought would need six, and had paid them for a day of standing about. He had guessed his crop by walking the rows and looking. That guess sets how many pickers he hires, how much transport he books, what price he can hold out for, and what credit he can raise against the season. He had no way to make it a number.

India grows more mango than any other country, around 22 million tonnes in 2023 to 2024 [7], and 86% of the country's farm holdings are under two hectares, working between them a little under half the cultivated area [14]. Most of the people growing this crop are therefore working at a scale where the fruit count per tree is the number everything else is planned around, and almost none of them have it.

Machine counting of fruit looks like a solved problem, and in one narrow sense it is. The published MangoYOLO detector reaches an F1 score of 0.968 at finding mangoes in an orchard image [3]. The difficulty is that a mango tree hides most of its own crop. When Wang, Walsh and Koirala counted fruit from a vehicle imaging both sides of a row and then harvested the trees to check, their imaging recovered 40.2% of the fruit that were actually there [4]. Detection is not the bottleneck. The bottleneck is the fruit no camera ever sees.

The standard repair is to multiply the visible count by a correction factor fitted on a few harvested trees. Across orchards that factor runs from 1.05 to 2.43 [3, 4], and the authors of that work say plainly that the final estimate is sensitive to it [4]. A number that varies by a factor of two across orchards is being treated as a constant of the orchard.

1.1 Aim

My aim was to test whether the total number of mangoes on a tree, including fruit no camera position can see, can be estimated from the images themselves rather than from a fitted constant, and whether that is measurably closer to the true count than the multiplier the field uses now.

1.2 Objectives

1. **A canopy simulator with a known answer.** Generate trees whose fruit count is known by construction, with occlusion following from the physics of light through foliage. *Done when* the visibility model reproduces the transmittance law it is built from, and the fraction of fruit it leaves visible sits inside the range real mango canopies show.
2. **An estimator that works from detection histories.** Recover the total from which viewpoints did and did not see each fruit. *Done when* it returns an estimate and an interval.

3. **Geometric recovery of the unobservable.** Use the reconstructed canopy to account for volume no viewpoint could see into. *Done when* it beats the estimator of objective 2 on canopies dense enough to hide fruit from every camera position.
4. **A fair comparison with current practice.** *Done when* every method is scored on identical trees from identical detections, so only the estimator differs.
5. **Validation on real trees.** Scan Alphonso trees, predict, then pick each tree and count. *Done when* the error against picked ground truth is measured and reported, whatever it is.

Objectives 1 to 4 are met and are reported here. Objective 5 is the work of the coming season.

1.3 How the work was organised

I ran the project in stages, each one ending in something I could check, and I fixed the order before I started for a reason. The evaluation had to exist before the estimator did. If I built the method first and the scoring afterwards, every choice about how to score it would have been made by someone who already knew which answer he wanted.

Stage	What it involved	What it depended on
1. Scope	The aim as one sentence, five objectives each with a *done when* test, and the success conditions written down and not edited afterwards.	Nothing. Deliberately first.
2. Canopy simulator	Tree geometry, foliage as a fixed grid of cells, trunk and limbs, and ray marching for every line of sight.	Stage 1, so that what counted as success was settled before anything could be tuned toward it.
3. Baseline estimators	Counting what is seen, the fitted multiplier that published work uses, and the classical capture-recapture family.	Stage 2, for trees with a known answer to score against.
4. Comparison protocol	Scoring every method on identical trees from identical detections, with the viewpoint count as the axis.	Stage 3.
5. Reconstruction	Recovering which parts of the canopy the cameras could see into, and which they never reached.	Stage 2.

Stage	What it involved	What it depended on
6. The estimator	Separating the detector's hit rate from the number of chances it had, and carrying measured fruit density into the volume no camera reached.	Stages 4 and 5.
7. Calibration	Checking the simulated trees against published counts from a real harvested orchard.	Stage 2. Ran late, and should not have.
8. Uncertainty	Prediction intervals, and a width correction that was tested and found wanting.	Stage 6.
9. Write-up and verification	Figures redrawn from the metrics file, every number cross-checked against it, and a defect log kept throughout.	Everything above.
10. Detector on real images *(scheduled)*	Replacing a modelled detector with a measured one, on published orchard photographs.	Stage 6.
11. Orchard campaign *(scheduled)*	Scanning trees, recording a prediction and an interval before touching the fruit, then picking each tree and counting every mango on it.	Stage 10, and the fruiting season.
12. Final analysis *(scheduled)*	Measuring the error against picked ground truth and reporting it, whatever it is.	Stage 11.

Two things about that order are worth stating rather than leaving to be inferred.

One stage ran later than it should have. Stage 7 checked my simulated trees against measurements somebody else had made on a real orchard, and I ran it after building the estimator rather than before. It found that my trees had no trunks or branches in them and were far easier than a real canopy, which meant a set of numbers had to be regenerated and several paragraphs rewritten. Everything internal had agreed with itself, because my checks and my simulator shared the same assumptions. Putting stage 7 immediately after stage 2 would have caught it in an afternoon. That is the single change I would make to the plan.

The plan changed once, on evidence. The project began around recovering fruit that no camera position can see. Once the simulator's physics was corrected, a full walk around a tree left only about 2% of fruit unseen from anywhere, so at that protocol there was little to recover and the premise did not hold. I narrowed the aim to the question the measurement supported, which is how few viewpoints an estimator needs to match what plain counting achieves with many, rather than adjusting the simulator until the original premise survived. The amendment is recorded in the project definition alongside the original wording, which I did not edit.

Stages 10 to 12 have not been done. They are scheduled rather than aspirational: the orchard campaign is timed to the Alphonso fruiting season between February and May, a cooperating grower in Ratnagiri has agreed to host it, and the protocol is already written down. Nothing in this report rests on them, and objective 5 stays open until they are finished.

2. Background

Three lines of work meet at the question I am asking, and the gap sits where they meet.

Detection is solved; counting is not. Convolutional detectors reached orchard-grade accuracy on fruit around 2019. MangoYOLO reports an F1 of 0.968 and average precision of 0.983 on a held-out test set, at 8 ms per image tile [3]. Later work moved from single images to video, tracking fruit between frames with a Kalman filter and the Hungarian algorithm so that the same mango is not counted twice [4]. The same shift had already happened for apples and mangoes in orchard imagery more broadly [1]. Both papers measure detection against fruit visible in the frame. Neither claims that number is the crop.

The gap between what is seen and what is there is large, and it is measured. The same group harvested twenty-one trees and counted every fruit by hand, which is the only ground truth that settles the question. Dual-view imaging recovered 40.2% of the harvest count and video tracking 62.3% [4]. That is the size of the problem: on the harder protocol, three fruit in five are never seen at all.

The repair is a constant, and the field knows it is not constant. Both papers apply a correction factor, the ratio of harvest count to machine count, fitted on harvested trees. Koirala and colleagues report factors from 1.05 for open canopies with sparse foliage up to 2.43 for large trees with dense foliage [3, 4]. Wang and colleagues state that the final orchard estimate is sensitive to that factor, and that the approach suits orchards managed so that all fruit sit on the outer wall of the canopy [4]. That is a candid description of a limitation, and it is the opening this project works in.

The statistics for this problem already exist, in another field. Ecologists have counted animals that hide for a century. Capture-recapture estimates a population from the pattern of which survey occasions saw which individuals: an individual seen on two occasions out of twelve is evidence that individuals like it are easy to miss, and the frequency distribution of sightings implies how many were missed entirely [13, 2, 8]. Horvitz and Thompson's estimator weights each observed individual by the reciprocal of its probability of being observed, so a fruit that had a one-in-four chance of appearing stands for four fruit [5]. None of this has been applied to fruit counting, as far as I can find, and the reason it fits is that a camera walked around a tree produces exactly the structure these methods need: repeated survey occasions of the same population.

The gap. Detection is solved. The hidden fraction is large, varies by a factor of two across orchards, and is currently handled by a constant. A body of statistics exists for populations that hide, and it has one blind spot that matters here, which the next section sets out. Nobody has brought the two together.

3. Method

3.1 The approaches I weighed

Approach	For	Against	Verdict
A person walks the rows and counts	No equipment	Hours per tree, and it counts only visible fruit anyway, less consistently than a detector	This is the problem, not a solution
Detect fruit in images, multiply by a fitted constant	Current published practice, cheap, fast	The constant varies from 1.05 to 2.43 across orchards and needs harvested trees to fit	The baseline I measure against
Reconstruct the tree in three dimensions, count the fruit visible in the reconstruction	Removes double counting between frames, and gives fruit size	Still counts only what was seen	A component, not an answer
Reconstruct, then estimate abundance statistically	Each viewpoint is a survey occasion; the canopy says what was never observable	More machinery, and it rests on a detection model that must itself be checked	Chosen

I chose the last because it is the only one that uses the information already sitting unused in the images. A fruit seen from three viewpoints out of twelve carries information that a fruit seen from all twelve does not, and counting throws that away.

3.2 Why the statistics work, and where they fail

Suppose every fruit on a tree has some chance of being caught by a camera on each pass. Fruit deep in the canopy have a low chance, fruit on the outside a high one. The distribution of how often fruit were actually seen then carries the information needed to estimate how many were never seen. This is capture-recapture, and on its own it is enough for animals that move.

It has one blind spot, and on a mango tree the blind spot is the whole problem. A fruit with no chance at all of being seen from any camera position leaves no trace in the data. It is not under-represented; it is absent. No amount of statistics recovers a population that left no evidence.

What closes the blind spot is the shape of the tree. This is the idea the project turns on.

3.3 What a reconstruction actually gives

Photogrammetry does not measure how dense a canopy is. It measures where the camera could see, by carving out the space along every unobstructed line of sight. That divides the canopy into three parts. Some volume was seen through by at least one camera. Some volume is where a line of sight stopped, so there is something solid there. And some volume was never reached by any line of sight from anywhere, so nothing at all is known about it.

That third part is what makes the hidden fruit estimable. Its *volume* is measured rather than assumed. Figure 1 shows it: a vertical slice through one simulated tree, with the region no camera reached picked out in charcoal. From two viewpoints it is a tenth of the canopy, and the wedge behind the trunk is plainly visible.

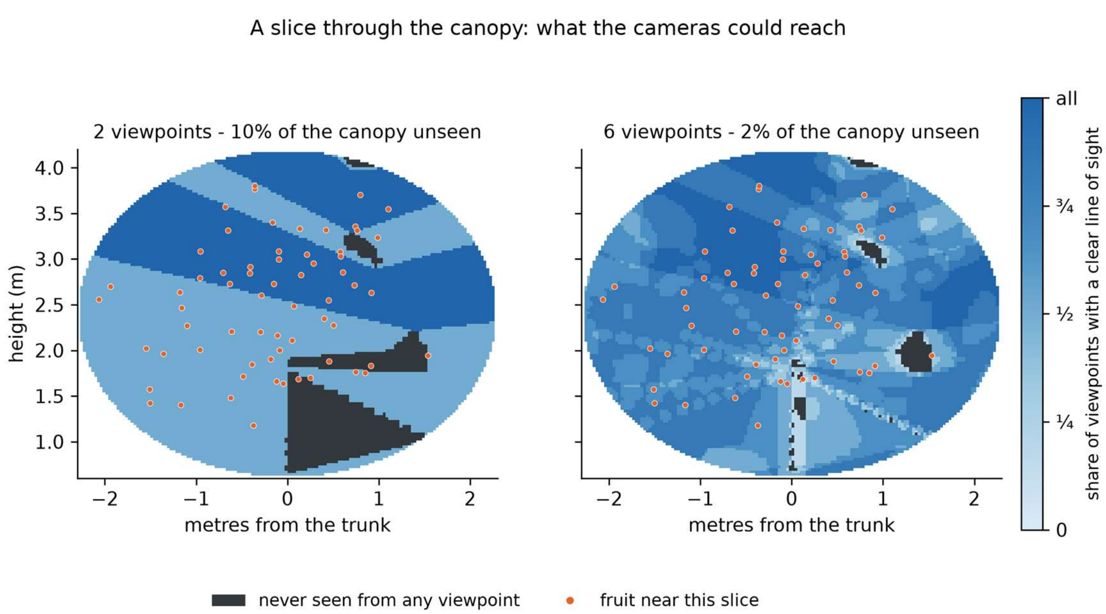

Figure 1. A slice through one simulated canopy. Shading gives the share of viewpoints with a clear line of sight; charcoal marks volume no viewpoint reached. Orange dots are fruit lying near the slice.

3.4 The estimator

The estimator works in three steps.

First, the reconstruction says how many camera positions had a clear line to each detected fruit. Together with the record of which positions actually detected it, that separates two quantities that are otherwise tangled: how many chances the detector had, and how good the detector is. Both become measurable, because the geometry supplies the first and the detection record supplies the second.

Second, each detected fruit is weighted by the reciprocal of its chance of appearing at all. A fruit that had one chance in four of being detected stands for four fruit. This recovers the fruit that could have been seen and were missed.

Third, the fruit that could never have been seen are estimated from the volume they occupy. The fruit density measured in each shell of the canopy is carried into the unobserved part of that same shell. Working shell by shell matters, because mangoes hang toward the outside of a canopy while the unobserved region is its middle, so a single density for the whole tree would put far too many fruit in the interior. Assuming instead that the seen and unseen parts of a given depth hold fruit alike is a much weaker claim, and it is the one the geometry supports.

3.5 Simulating trees, and why

An estimator has to be judged against a truth, and on a real tree the truth costs a harvest to obtain. So the development ran on simulated trees whose count is known by construction, and the real orchard is being kept for the final test rather than spent on debugging.

That only works if the simulator is not quietly easier than a real tree, which is a trap I fell into and had to climb out of twice. Section 4 sets out how.

The canopy is an ellipsoid of foliage with a leaf area density, a trunk, seven scaffold limbs, and fruit placed with a bias toward the outside. Foliage is built once per tree as a grid of cells that are either leaf or air, with the occupancy set so that average transmittance matches the Beer-Lambert law, and every line of sight is traced through that one fixed arrangement.

Whether a fruit is found is not a yes or no. A detector sees a fruit rather than a point, and how much of that fruit is showing decides whether it is found, which is most of the difference between the F1 near 0.97 that published mango detectors reach on curated test images and the 0.89 they reach on daytime orchard images [3]. So visibility is measured over nine points spread across the face each fruit presents to the camera, and recall climbs with the share showing. Appendix C gives the mathematics.

4. Evaluation design

The design of this study is mostly a record of things I got wrong, so it is easier to tell it that way.

4.1 The first simulator flattered the answer

My first version drew each viewpoint's occlusion independently, with the transmittance along that line of sight as the probability. It reported 86% of fruit visible, every estimator landed within a few percent of the truth, and counting what was visible was only 13.6% low. The problem looked answered before it had been asked.

The error is that leaves do not rearrange themselves between viewpoints. Occlusion is fixed by the geometry and strongly correlated between neighbouring camera positions, so independent draws turn a fruit with a 30% chance per view into one that is near-certain to be seen across twelve. A fruit buried in the canopy interior, which should be invisible from every direction, became almost certain to be found.

The fix was to build the foliage once per tree and trace every line of sight through that one arrangement. Two further defects surfaced immediately afterwards: foliage scattered as a uniform mist rather than clumped into shoots, and a ray-marching step coarser than the cells it was meant to test, so thin foliage was being stepped over. Both are logged with cause, fix and verification in the repository.

4.2 The premise was wrong, and the corrected simulator said so

With the physics fixed, twelve viewpoints left only about 2% of fruit never seen. I had framed the project around recovering fruit that no camera can see, and at that protocol there are few of them.

Walking a full circle around a tree recovers most of what one view hides, because a fruit hidden from one side is usually exposed from another. The severe occlusion is at one and two viewpoints, where 40% and 15% of fruit are never seen. That is also the protocol anyone scanning a real orchard would use, because nobody walks twelve camera positions around each of two hundred trees.

I restated the aim rather than adjusting the simulator to rescue it. The question became how few viewpoints an estimator needs to match what plain counting achieves with many.

4.3 Checking the simulator against the field, not against itself

A simulator tuned until it agrees with its author proves nothing. The check that matters is against measurements someone else made on real trees.

Wang, Walsh and Koirala recovered 40.2% of harvest count from dual-view imaging [4]. My simulator, at the time, said 85% of fruit were visible from two viewpoints. The trees were far too easy.

The diagnosis came from asking what would have to change to close the gap. Raising leaf density until the visible fraction fell that far would have required a leaf area index above 20, against the 3 to 6 a bearing mango actually carries [9]. So the missing occluder could not be foliage. It was wood: the trees had no trunk and no limbs, and a mature mango carries a trunk near 280 mm across that blocks a line of sight completely.

Adding the wood and resetting leaf density to physical values brought one viewpoint to 57% visible, which brackets the published dual-view figure acceptably. Two viewpoints remained at 82% against their 62%, so the simulator was still easier than the field.

The rest of that gap closed when detection stopped being a yes or no. Once recall depended on how much of a fruit was showing, sweeping the detector's ceiling and its threshold found a setting that reproduces both published figures at once: 37% of fruit found from one viewpoint against their 40.2% from dual view, and 61% from two viewpoints against their 62.3% from video tracking. The correction factors my trees then demand, from 2.99 at one viewpoint to 1.04 at twelve, span the 1.05 to 2.43 measured across real orchards. Matching three separate published quantities from one setting is closer agreement than I expected.

I am reporting that as a calibration of the whole pipeline rather than as a measurement of a detector, because the two are not the same thing. Their figure counts detections against a harvest, so it includes fruit no camera was ever pointed at, high in the tree or on the far side of a row, while my cameras see the whole tree. The calibrated setting may be standing in for losses this model does not represent. That is why the study re-runs its comparison under a better detector and a worse one, so the conclusion can be checked against the setting rather than resting on it. Figure 2 puts the simulator's visible and never-visible fractions against viewpoint count, with both published measurements marked.

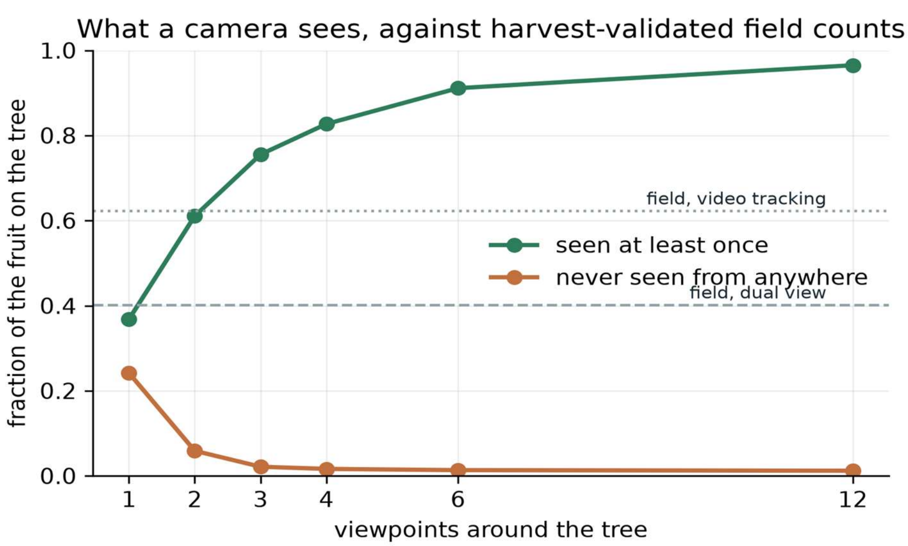

Figure 2. Fruit visible and fruit never visible, against viewpoint count, with the two published field measurements marked.

4.4 The protocol

A hundred and thirty trees were drawn with canopy dimensions, leaf area density and fruit load varying across physically defensible ranges, and split three ways. Twenty fit the multiplier, exactly as the field fits it. Sixty were held back for scoring, and every number in the next section comes from those sixty. Fifty more were drawn solely to calibrate the width of the prediction intervals, so that the trees the coverage is reported on never inform how wide the intervals are. Each tree was scanned at every viewpoint count, so the viewpoint curve is measured within tree rather than across different trees. Every estimator saw identical detections from identical trees, so the only thing differing between them is the estimator.

The detector sensitivity in Section 5.6 is a separate experiment with its own trees, thirty per detector setting, of which the first ten fit that setting's multiplier and the remaining twenty are scored.

One structural guard is worth naming. The scan hands an estimator only quantities a reconstruction would recover, so the true canopy cannot reach an estimator even by accident, and a test asserts it. Making the mistake impossible is worth more than intending not to make it. The detector re-run under a better and a worse detector uses thirty fresh trees per setting, ten fitting that setting's multiplier and twenty scored.

5. Results

Every number below comes from `run_simulation_study.py` at seed 20260903, on sixty held-out simulated trees. No number here was measured on a real tree.

5.1 How much a camera misses

From a single viewpoint, 24% of fruit have no clear line of sight from anywhere, and once the detector's own misses are added only 37% are found at all. Two viewpoints find 61%, three find 76%, and twelve find 97%. Counting what is found is therefore 63.2% low at one viewpoint and 3.5% low at twelve, which is the shape of the problem in two numbers.

A second check falls out of this. The correction factor needed to repair the visible count, fitted the way the field fits it, runs from 2.99 at one viewpoint down to 1.04 at twelve. Published factors measured across real orchards run from 1.05 to 2.43. My simulated trees demand corrections of the same size as real ones, which is reassuring about the trees rather than about the estimator.

5.2 Accuracy against effort

Viewpoints	Count what is seen	Fitted multiplier	Capture-recapture	**Reconstruction**
1	63.2%	18.3%	not estimable	not estimable
2	38.9%	11.1%	15.9%	**9.5%**
3	24.5%	7.9%	3.8%	**3.5%**
4	17.3%	6.4%	3.4%	**2.9%**
6	8.9%	3.8%	3.1%	**1.8%**
12	3.5%	1.9%	3.0%	**1.2%**

Mean absolute error in total fruit count.

The reconstruction estimator is the most accurate at every viewpoint count where it is defined. Its margin over the fitted multiplier is largest in the middle of the range, at 2.3 times better on three viewpoints and 2.1 times on six, and narrows at both ends. At two viewpoints it is only 1.2 times better, because the detection histories are then too short to say much. At twelve there is little left hidden for any method to recover. Figure 3 plots error against viewpoint count for every estimator.

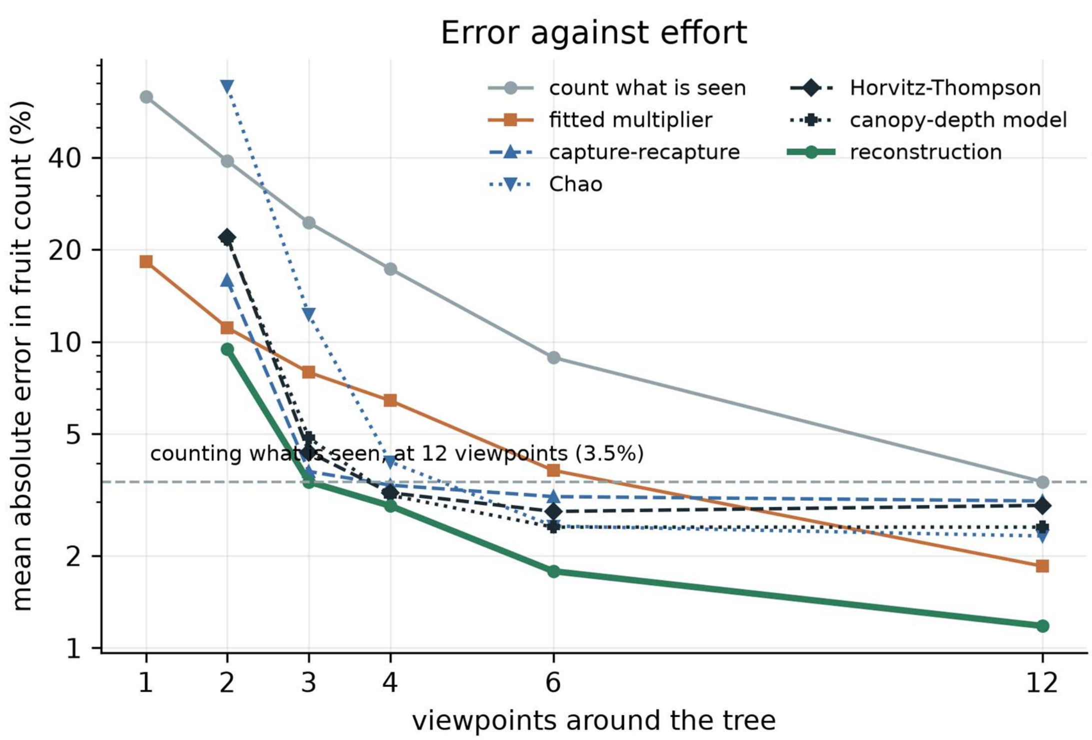

Figure 3. Error against viewpoint count for every estimator. The dashed line is plain counting at twelve viewpoints.

An average error can hide two different failures. A method can miss by a little on every tree, or be right on most and badly wrong on a few, and a grower cares about the second far more than the first. Figure 4 puts the per-tree estimates against the truth at three viewpoints. The multiplier's points fan out as the fruit load rises, which is what a single constant does when the hidden fraction is not constant. The reconstruction estimator's points stay near the line across the whole range.

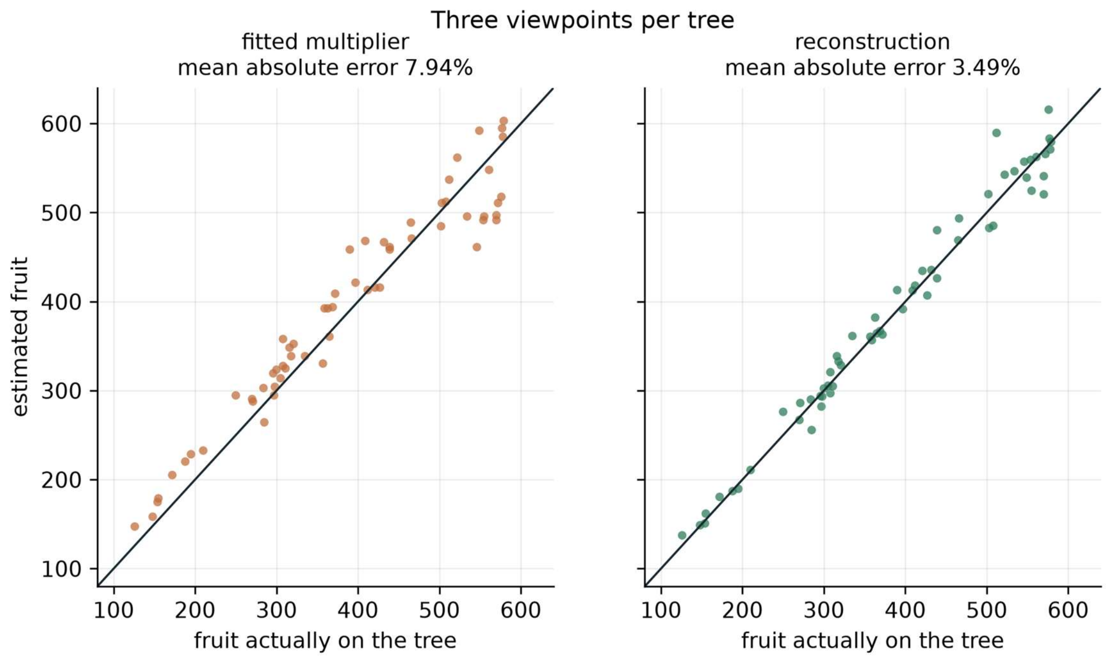

Figure 4. Estimated against actual fruit for sixty trees at three viewpoints. The line is exact agreement.

5.3 The same accuracy for less walking

Plain counting at twelve viewpoints reaches 3.5%. The reconstruction estimator reaches 3.5% at three viewpoints and 2.9% at four, so three viewpoints with it match twelve without, and four beat them. The fitted multiplier never gets there: at twelve viewpoints it reaches 1.9%, but it needs all twelve to do it. For a grower with two hundred trees that is the difference between a morning and a day. Figure 5 shows how few viewpoints each method needs to reach that line.

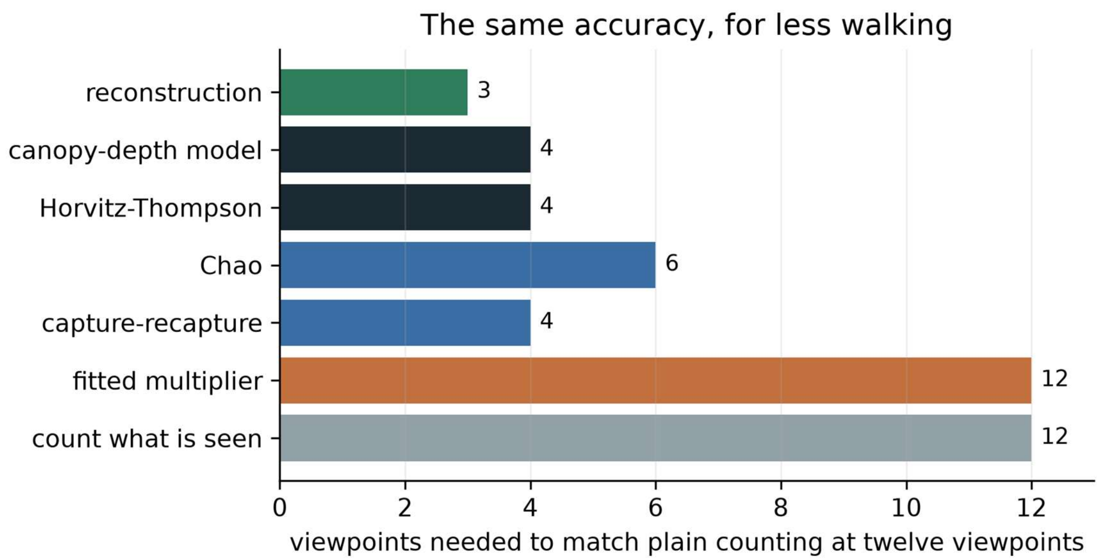

Figure 5. Viewpoints each method needs to match plain counting at twelve.

5.4 A single viewpoint is refused, not guessed

From one viewpoint every observed fruit has been seen exactly once, so the sighting distribution is a single point and the detection probability is not identified. My first version returned numbers anyway, with errors of 331% and, for one estimator, 6577%. The estimators now refuse. An estimator that declines is more useful than one that guesses, and the refusal is itself a result: a single photograph cannot support this family of methods at all, and the only thing available there is a correction factor of about three, bought with a harvest.

5.5 Where the correction earns its keep

Canopy	Viewpoints	Found	Count what is seen	Fitted multiplier	Reconstruction
Dense	2	57%	43.3%	11.3%	**10.2%**
Dense	3	73%	26.6%	7.0%	**4.5%**
Dense	12	95%	4.8%	1.9%	**1.4%**
Open	3	78%	22.2%	9.4%	**3.8%**
Open	12	98%	2.3%	2.3%	**1.0%**

The advantage holds across canopy density rather than depending on it. What changes with density is how much there is to recover: a dense canopy at two viewpoints hides 43% of its crop from a plain count, an open one at twelve hides 2%. Figure 6 splits the same result by canopy density and viewpoint count.

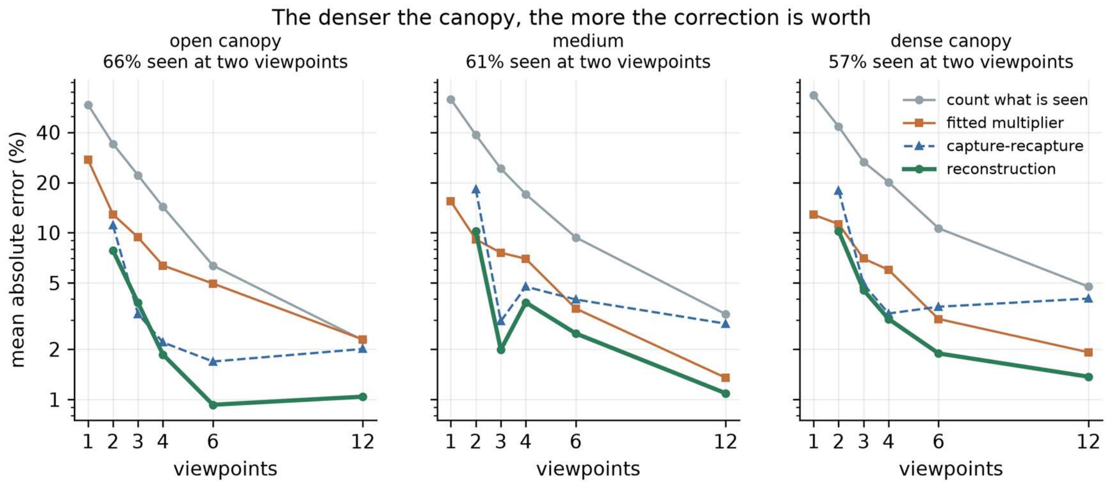

Figure 6. Error by canopy density and viewpoint count.

5.6 The estimator depends on the detector, and there is a point where it fails

The detector is the one part of the pipeline this project does not build, so the conclusion is re-run under a better and a worse one. The result is a boundary rather than a reassurance.

Detector	Viewpoints	Found	Count what is seen	Fitted multiplier	Reconstruction
Calibrated	2	60%	40.5%	11.0%	**8.0%**
Calibrated	3	74%	26.1%	8.7%	**3.8%**
Calibrated	6	91%	9.1%	2.4%	**1.8%**
Better	2	82%	17.9%	6.1%	**3.4%**
Better	3	92%	8.3%	3.5%	**1.7%**
Worse	2	41%	58.9%	**12.5%**	21.9%
Worse	3	55%	45.5%	11.9%	**7.3%**
Worse	6	78%	21.7%	5.2%	**3.4%**

With a poor detector and only two viewpoints the estimator loses, and loses badly: 21.9% against the multiplier's 12.5%. That combination leaves it fitting a detection model to histories that are both short and mostly empty, and a wrong model applied confidently is worse than a blunt constant applied cautiously. Three viewpoints are enough to recover the advantage even with the poor detector, and six restore it fully.

This is the practical limit of the method as it stands, and it is a specific and checkable one: do not use it on two viewpoints unless the detector is known to be good. Figure 7 shows the comparison under all three detectors.

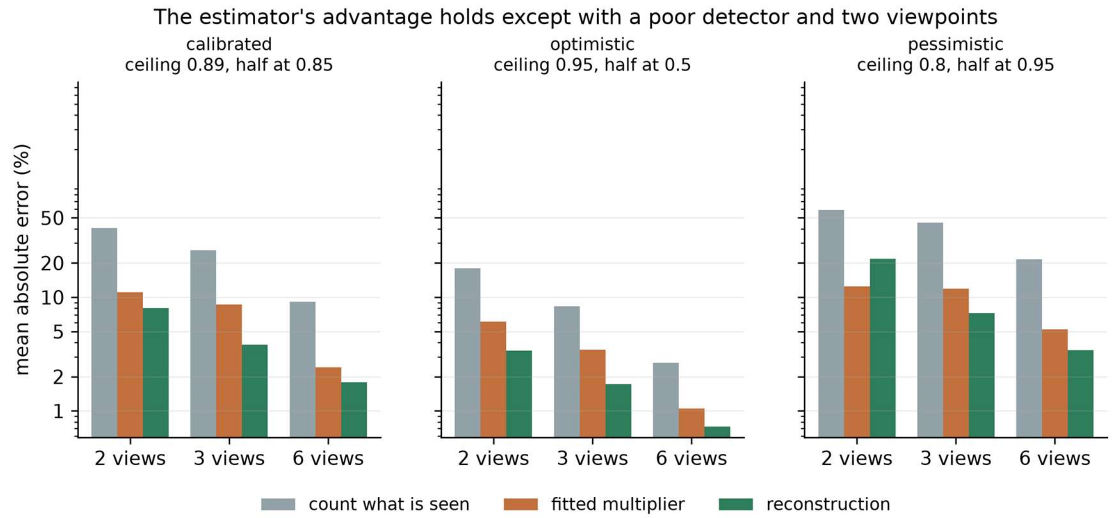

Figure 7. The comparison under three detectors. The estimator's advantage holds everywhere except the poor detector at two viewpoints.

5.7 The intervals, and a correction that did not work

Raw prediction intervals from the parametric bootstrap cover 88% of trees at two viewpoints against a claimed 90%, which is close, and then drift upward to 100% at four viewpoints and above, which is conservative rather than wrong but is still not what the interval says.

I rescaled them by a multiplier fitted on fifty trees held back for that purpose, and the result is mixed. At three, six and twelve viewpoints the coverage lands where it should, at 90%, 93% and 92%, and the interval narrows from 23% of the count to 17% at three viewpoints and from 17% to 5% at twelve. At four viewpoints the correction overshoots and coverage falls to 80%, below what it claims. At two viewpoints it takes an already reasonable 88% and pushes it to 100%.

My first explanation was that the multiplier was fitted on too few trees. That turned out to be wrong, and the way I know is that I tested it: tripling the calibration set from twenty trees to fifty moved the multipliers by almost nothing, from 1.259 to 1.309 at two viewpoints and from 0.511 to 0.509 at four. A quantity that does not move when its sample triples is not the noisy part. The correction is the wrong shape rather than imprecise, because one number per viewpoint count cannot repair an interval that is miscalibrated by different amounts on different trees. Making it depend on how much of each canopy went unobserved is the next thing to try.

Both series are reported for that reason, and a grower using this today should use the uncalibrated interval, which is conservative everywhere and close to correct at the protocols that matter. Figure 8 shows coverage and width before and after the correction, against the 90% claimed.

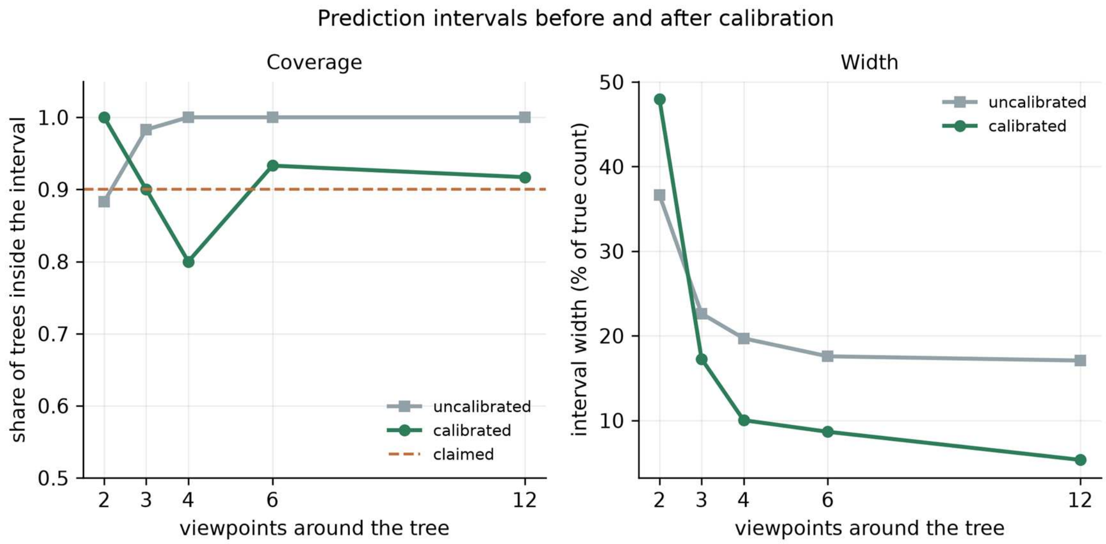

Figure 8. Coverage and width, before and after calibration, against the 90% claimed.

5.8 The objectives, revisited

The simulator reproduces the transmittance law it is built from, checked against the closed form in the test suite. Its fruit-found fractions now bracket both published field figures, and the correction factors it demands span the range measured across real orchards. The estimator returns a total and an interval. Using the reconstruction beats the smooth canopy-depth proxy it replaced at every viewpoint count, by more than a factor of two at two viewpoints. Every method was scored on identical trees from identical detections.

The fifth objective, validation against picked and counted trees, is not met. It cannot be met before the fruit exists. The orchard campaign is scheduled for February to May 2027, and the result will be reported regardless of what it shows.

6. Discussion

6.1 The answer to the question I asked

The hidden fruit can be estimated from the images themselves. On simulated trees the estimate is about twice as accurate as the fitted multiplier the field uses across most of the range, and needs no harvested trees to calibrate against, which is the practical difference: a multiplier has to be bought with a harvest, and this does not.

It is not uniformly better. On two viewpoints with a poor detector the multiplier wins, and by a wide margin. The method needs enough sightings to fit a detection model, and where it does not have them a confident wrong model is worse than a blunt constant.

The claim is bounded in a specific way. It is measured on simulated trees whose occlusion the previous section shows to be milder than a real hedgerow. It is not yet a claim about Alphonso.

6.2 What produced the result

Two decisions did most of the work. The first was to keep the reconstruction's measurement of unobserved volume rather than a smooth proxy for canopy depth. The proxy assumed foliage was spread evenly, which is the assumption clumped foliage breaks, and it barely improved on plain capture-recapture. Replacing it roughly halved the error at two viewpoints.

The second was separating the detector's hit rate from the number of chances it had. Tangled together they are not identifiable; split between the geometry and the detection record they both are. Chao's estimator, which uses the sighting distribution alone, reaches 68.2% error at two viewpoints where the split estimator reaches 9.5%.

6.3 Where it sits against the field

Wang, Walsh and Koirala report a bias-corrected error of 18.0 fruit per tree against a mean of 156.5, which is 11.5% [4]. I have not set my figures beside that number and will not, because theirs is measured on real trees under a harder protocol and mine is simulated. The two are not comparable, and putting them next to each other would imply a comparison the work does not support. The comparison this project can make is against the fitted multiplier on identical simulated trees, and that is the one reported.

What can be said against the literature is narrower and still worth saying. The correction factor the field relies on varies from 1.05 to 2.43 across orchards, and the authors note the final estimate is sensitive to it [3, 4]. An estimator that reads the hidden fraction off each individual tree removes that dependence in principle. Whether it does so in practice is a harvest away.

6.4 Creative decisions, named

Two things in this project were choices rather than steps.

Treating fruit counting as a wildlife abundance problem is the first. The methods ecologists use for animals that hide map onto a camera walked around a tree with no forcing, because both produce repeated survey occasions of one population, and I have not found the transfer made before.

Using the reconstruction for what it measures rather than what it appears to measure is the second. A three-dimensional reconstruction looks like a tool for finding where things are. Its more useful output here is a map of where the cameras could not look, which is a statement about absence rather than presence.

6.5 Limitations

The results are simulated. The simulator is calibrated against published field measurements rather than validated against my own, and it represents a free-standing tree scanned on foot rather than a hedgerow imaged from a vehicle.

The intervals are calibrated on fifty trees, which is enough for the correction to be stable but not for it to be the right shape, and the correction misses at two of the five viewpoint counts.

Fruit detection is modelled rather than performed. The model is anchored to published detector performance, and the conclusion is checked against a better and a worse detector, but no detector was run on an image in this work.

Fruit positions are assumed recoverable in three dimensions, which photogrammetry does under good conditions and less well in wind or poor light. That assumption is untested here and is the one I would attack first.

7. Ethics and responsible use

A wrong count costs money, and the two errors are not equal. An over-estimate means a grower hires pickers who stand about, which is the mistake that started this project. An under-estimate means fruit left on the tree past ripeness. The estimator reports an interval alongside the number for this reason, and the interval is currently conservative, which errs toward telling a grower they know less than they might. That is the direction I would rather err in.

No personal data. The system photographs trees. Where a person appears incidentally in an orchard video, frames are used for geometry and not retained beyond the reconstruction.

Whose orchard, and whose data. The orchard campaign will run on a cooperating grower's trees and will destroy part of a harvest, since the ground truth is obtained by picking. That cost falls on him, so it will be agreed in writing beforehand, the fruit will be his to sell, and the trees will be his choice. The data from his orchard is his, and the published dataset will identify the region and not the holding.

A tool that serves some growers better than others. Section 5.5 shows the accuracy varies with canopy density, and dense unpruned canopies are the harder case. Unpruned canopies belong disproportionately to growers with less capital to spend on management. A tool that works best on well-managed orchards would widen a gap it claims to close, so accuracy is reported by canopy density rather than as a single average that would hide it.

Dual use. A buyer with a good crop estimate and a grower without one is a worse bargaining position than the one the grower has now. The intended deployment is in the grower's hands.

8. Reflection and next steps

What I learned. The technical lesson was that a simulator is a hypothesis about the world and has to be tested against the world, not against itself. My trees were missing their trunks for two days and I did not notice, because everything internally agreed. What caught it was a number from someone else's harvested orchard.

The habit that mattered more was reading a discouraging result as information. Twice my own measurement contradicted what I had set out to show: once when the corrected simulator said there were barely any invisible fruit at twelve viewpoints, and once when the field data said my trees were too easy. Both times the useful move was to change the claim rather than the code.

What I would do differently. I would have gone to the published field measurements before building the simulator rather than after. The 40.2% figure was available the whole time and would have caught the missing wood on the first day. I built inward from physics I trusted instead of outward from measurements someone had already made.

Next steps, with what each needs. A fruit detector run on public orchard images, to replace the assumed 0.95 reliability with a measured one. Interval width calibrated on simulated trees held back for that purpose, with coverage then reported on trees never used for it. The orchard campaign in February to May 2027: scan trees, predict with an interval recorded before picking, then pick each tree and count every fruit. And a test of the thing I cannot simulate honestly, which is whether photogrammetry recovers a mango canopy well enough in field conditions for any of this to hold.

Acknowledgements

The grower in Ratnagiri set this project in motion and will host the orchard campaign. He asked not to be named and had no involvement in the design or analysis. My father carried the conversation that started it and read the drafts of this report. The published measurements behind the calibration in Section 4 were made by researchers who harvested and counted twenty-one trees by hand, and this project leans on that work throughout.

A note on AI use

I used an AI assistant, Claude, throughout this project, and the use was substantial rather than incidental. This note sets out what it did and what I did, because the assessor is entitled to know which is which.

What the assistant did. It wrote most of the code in the repository from my direction, including the canopy simulator, the estimators and the analysis scripts. It drafted sections of this report from my notes and my decisions about what the report should argue. It found several of the defects recorded in the fix log, and proposed the parametric bootstrap that replaced my first attempt at prediction intervals.

What I did. I chose the problem and why it mattered, having watched the grower it affects. I set the aim, the objectives and the success conditions before any code existed, and I held to them when the results contradicted what I had expected. I decided that the aim should narrow rather than the simulator be adjusted when the corrected physics undercut my original framing, and I decided to stop tuning the simulator toward the published field figure rather than force agreement. I read every number against the metrics file, and I can explain the method, the statistics and the limitations of this work.

What I verified. I ran the code and the test suite myself. Every figure in this report is drawn from the same file the text quotes, and I checked the report's numbers against it line by line.

Where the boundary is. I am accountable for every claim here. Where I could not verify something, the report says so.

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Note on the reference list. Entries 2, 3, 4, 5, 8 and 13 carry the argument and I have read them. Every entry's bibliographic details have been checked against the original record. That check corrected two page ranges and replaced the source behind the production and smallholding figures, which had been attributed to a publication that does not carry them.

Appendix A: glossary

Term	Meaning
Occlusion	One object hiding another from a viewpoint
Detection history	The record of which viewpoints saw a given fruit
Capture-recapture	Estimating a population from how often individuals are re-sighted
Horvitz-Thompson estimator	Weighting each observed individual by the reciprocal of its chance of being observed
Leaf area index	Leaf area above a patch of ground, divided by that ground area
Beer-Lambert law	Light falls off exponentially with the amount of material it passes through

Term	Meaning
Space carving	Recovering shape by removing volume that a camera can see through
Mean absolute percentage error	Average size of the error, as a percentage of the true value
Coverage	The share of cases a stated interval actually contains

Appendix B: provenance and how to reproduce

- **Code.** `aamganak/` in the public repository at https://github.com/devadit1515/mangoguard. Seed 20260903 throughout.
- **Reproduce.** `python scripts/run_simulation_study.py` regenerates every number into `artifacts/sim_metrics.json`, in tens of minutes on a consumer laptop, CPU only; `python scripts/make_figures.py` redraws every figure. `pytest` runs twenty tests, including the closed-form check on the transmittance model. The detector settings are labelled optimistic and pessimistic in the code.
- **Population.** A hundred and thirty trees: twenty to fit the multiplier, sixty held out for scoring, fifty drawn only to calibrate interval width. A further thirty per detector setting for the sensitivity experiment, ten of those fitting that setting's multiplier. Canopy radius 1.8 to 2.8 m, half-height 1.4 to 2.2 m, leaf area density 0.8 to 2.0, fruit load 120 to 600. Fruit radial position is drawn from a Beta(3,2) distribution, a mean radius of 0.6 of the canopy, and the carved volume is estimated from 2,500 Monte Carlo points per tree.
- **Detector.** Recall rises with the share of each fruit showing, measured over nine points across the face it presents to the camera, saturating at a ceiling of 0.89 and halving at 0.85 showing. Those two values are calibrated so that the pipeline reproduces both published field detection rates, and the study re-runs its headline comparison under a better and a worse detector.
- **Record of defects.** `FIX_LOG.md`, thirteen entries with cause, fix and verification, four of them open.
- **Record of scope changes.** `PROJECT_DEFINITION.md`, amended by appending only.

Appendix C: the mathematics

Attenuation through foliage. A line of sight of length L through foliage of leaf area density a has an unobstructed probability $\exp(-GaL)$, with $G = 0.5$ for leaves pointing in every direction equally [6, 10]. The canopy is discretised into cells of side h, each foliage with probability $1 - \exp(-Gah)$, so that a ray crossing L/h cells is clear with probability $(1 - (1 - \exp(-Gah)))^{L/h} = \exp(-GaL)$, recovering the continuous law. Discretising changes the correlation between viewpoints, which is the point, and leaves the mean unchanged, which the test suite checks.

Clumping. Cell occupancy is drawn as a smoothed random field thresholded at the target occupancy, giving foliage correlated over roughly 20 cm. Canopy models carry a clumping index for this reason: scattered-leaf theory gets gap statistics wrong even when mean density is right [11].

Detection. Writing v for the share of a fruit's face that is showing from a given viewpoint, recall is $q\,\sigma(\kappa(v - v_{1/2}))$ for a logistic σ, with ceiling q, half-recall point $v_{1/2}$ and sharpness κ, and zero when nothing is showing. The estimator does not know v and treats each viewpoint as clear or not, so its model is deliberately coarser than the process generating the data, which is the situation any real deployment is in. A fruit with c clear viewpoints and hit rate q is then modelled as detected $y \sim \mathrm{Binomial}(c, q)$ times, entering the data only if $y \geq 1$. The hit rate is fitted by maximising the zero-truncated likelihood

$$\prod_i \frac{\binom{c_i}{y_i} q^{y_i}(1-q)^{c_i - y_i}}{1-(1-q)^{c_i}},$$

the truncation correcting for fruit absent from the sample by construction.

Total. With inclusion probability $\pi_i = 1-(1-q)^{c_i}$, the Horvitz-Thompson total over observable fruit is $\sum_i \pi_i^{-1}$ [5]. Writing V_b for the volume of canopy shell b and u_b for the share of it no viewpoint reached, the fruit density measured in that shell is $\rho_b = (\sum_{i\in b} \pi_i^{-1})/(V_b(1-u_b))$, and the estimate is $\widehat{N} = \sum_b \rho_b\, V_b$, over eight radial shells. Shells with no observable volume take ρ from a log-linear fit across the shells that have it.

Intervals. By parametric bootstrap with eighty replicates: a synthetic tree is drawn with $\widehat{N}$ fruit positioned by the fitted density profile, each inheriting the clear-view count of the canopy point it sits at and detected at rate q, and the estimator is re-run. Fruit that go undetected drop out as they do in the real data, so the truncation is reproduced rather than assumed away [12].

End of report.